

Intracellular Signalling Pathways

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06_Intracellular_signaling_pathways_2.pdf (19 slides) · Refs: Guyton & Hall 13th ed. (Unit II ch. 6); Ganong 25th ed. (Section I ch. 5)

Legend: ★ = high-yield · [added] = not on slides ·  = the lecturer's handwritten annotation. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

⚠ The lecturer **scribbled out** the "3 pathways to alter transcription" list (Slide 14). It is kept here with strikethrough and is likely **not examinable**; confirm with the lecturer.

🔗 **Bridge from Lecture 05 (Hormones & Receptors):** Lecture 05 ended with a tree of four ways a receptor turns a hormone signal into a cell effect (channels, G-protein, enzyme, transcription) and showed where each hormone class's receptor sits. This lecture **zooms into each branch** of that tree.

1. Title / Objectives

Objective	Likely tested as
★ Explain intracellular signalling pathways of hormone receptors	Transmembrane vs cytoplasmic; which hormones use which
★ Describe channel activity	Pentameric ligand-gated (ACh, GABA) vs tetrameric aquaporin (ADH)
★ Identify G-protein-coupled receptors	GDP/GTP switch; α , β , γ ; Gs / Gi / Gq ; 7 transmembrane loops
★ Enzyme activation and transcription	Type I (intrinsic, insulin) vs type II (associated, GH-JAK2); HRE

Flow: Pathway tree → transmembrane receptors → (A) ion channels → (B) GPCRs & heterotrimeric G-proteins → (C) enzyme-linked receptors → cytoplasmic/nuclear receptors & transcription → conclusion → case.

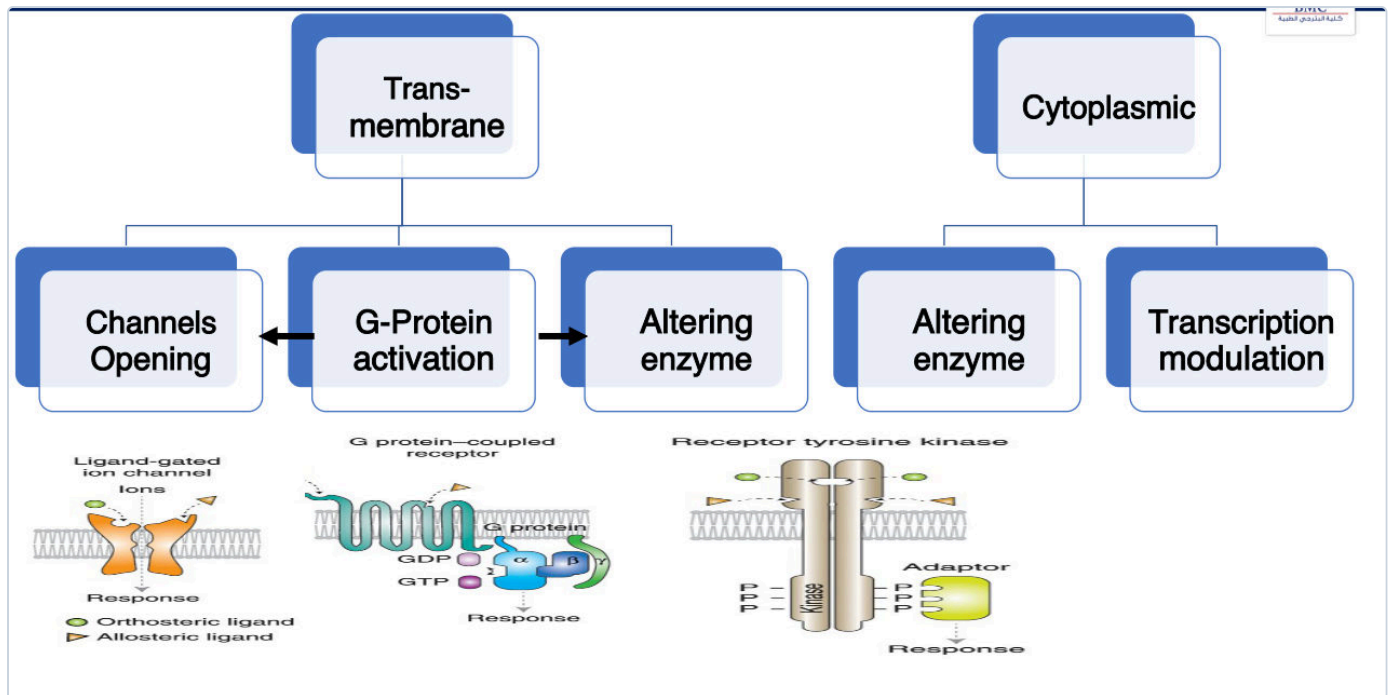
2. Main Content (Lecture Order)

2.1 The Pathway Tree (Slides 4-6), repeated from Lecture 05

Receptor location	Mechanism
★ Transmembrane	A. Ion channel opening · B. G-protein activation · C. Altering enzyme activity
★ Cytoplasmic	C. Altering enzyme activity · D. Transcription modulation

What signalling pathways provide:

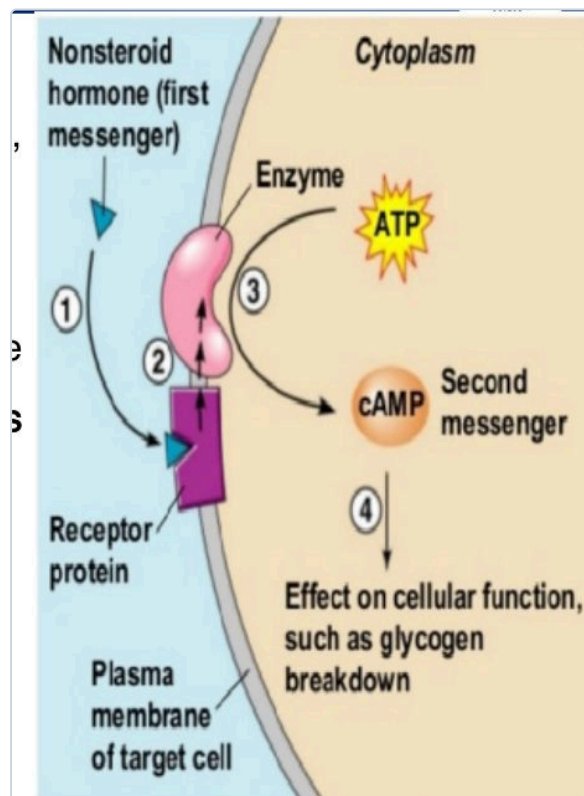
- ★ **Amplification** (same effect, bigger) and **distribution** (different effects)
- ★ **Regulation and feedback control**



Slide 5: Signalling pathway tree with example receptors

2.2 Transmembrane Receptors: General (Slide 7)

- ★ For **lipid-insoluble** hormones: **proteins & peptides, catecholamines, serotonin, melatonin**
- Located **in or on the cell-membrane surface**
- Hormone binding regulates **target proteins (enzymes or ion channels)** either:
 - ★ **Directly**, or
 - ★ **Through G-protein coupling**



Slide 7: Membrane receptor → enzyme → cAMP → effect

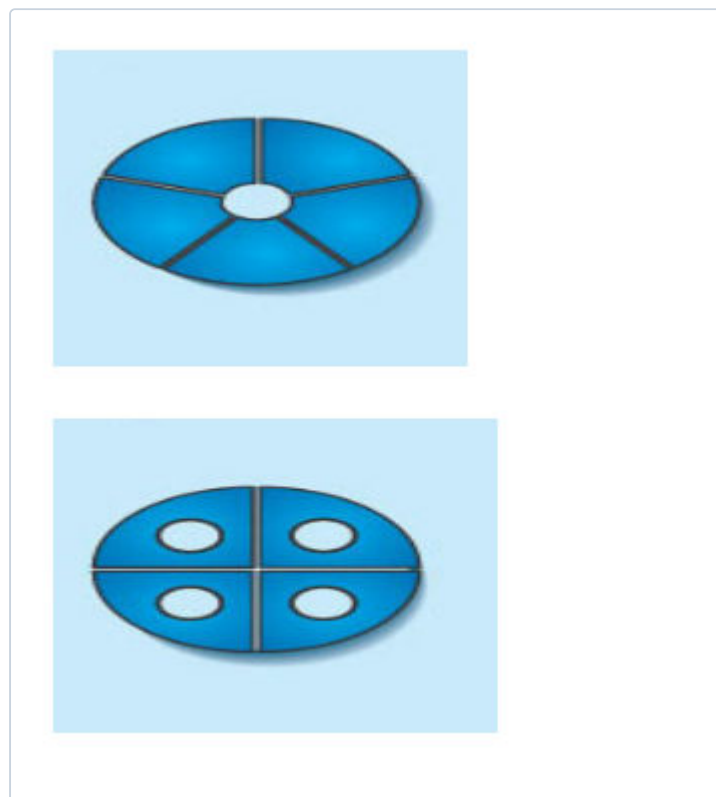
🔗 **Recap: Transmembrane receptors.** Water-soluble hormones (peptides, catecholamines, serotonin, melatonin) can't enter the cell, so they bind surface receptors that act on channels or enzymes, directly or via G-proteins.

🔗 **Link:** The simplest and fastest action is a receptor that *is* a channel.

▶ **Up next: (A) Ion channels.**

2.3 (A) Opening Ion Channels (Slide 8)

Channel	Structure	Pore	Ligand / example
★ Pentameric ligand-gated (📝 "ligand-gated" underlined)	5 identical or similar subunits	★ One single pore formed by all 5	★ Acetylcholine, GABA ; cation (K^+ on slide) or anion (Cl^-) channels
★ Aquaporin water channel (📝 underlined)	Tetramer (4 subunits)	★ One pore per subunit (4 pores)	★ ADH



Slide 8: Pentamer with one central pore (top) vs aquaporin tetramer with a pore in each subunit (bottom)

[added: the nicotinic ACh channel passes Na^+ inward mostly (and K^+ out), causing depolarisation; GABA-A opens Cl^- channels, causing inhibition. ADH (via V2 receptors and cAMP) inserts aquaporin-2 into collecting-duct membranes.]

🔗 **Recap: Channels.** Ligand-gated pentamer = 5 subunits, 1 pore (ACh, GABA). Aquaporin = tetramer, 4 pores (ADH).

🔗 **Link:** Most hormone receptors don't open a channel themselves; they work through a G-protein relay.

▶ **Up next: (B) G-protein-coupled receptors.**

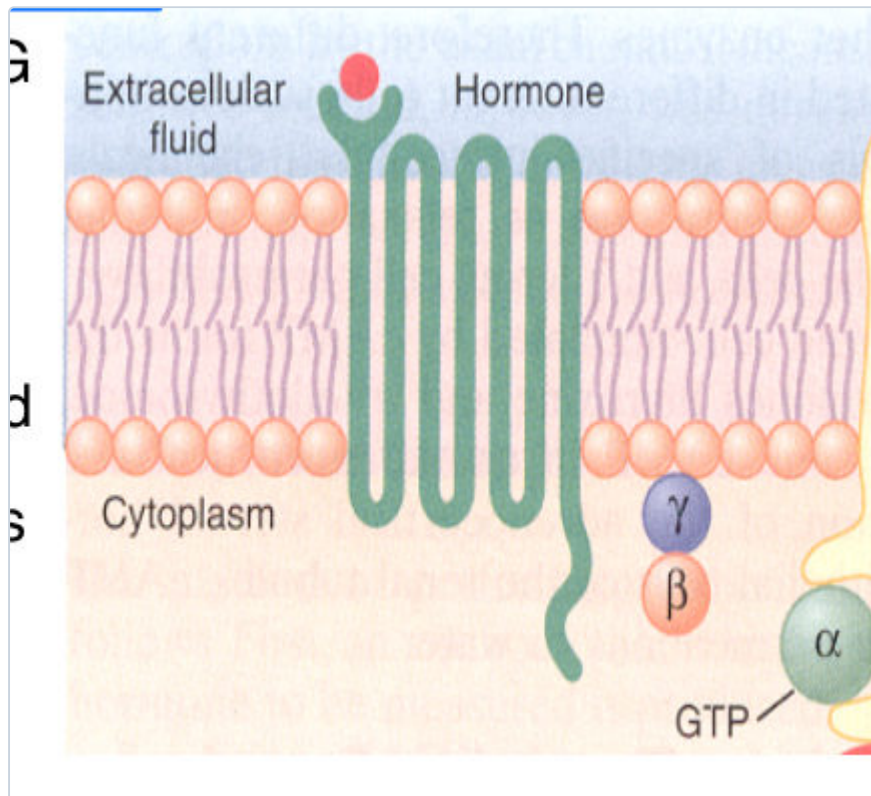
2.4 (B) G-Protein-Coupled Receptors (Slides 9-11)

★★ The G-protein switch (Slide 9):

1. Named for their ability to **bind guanosine nucleotides**
2. Activating signal reaches the GPCR → G-protein ★ **exchanges GDP for GTP**
3. ★ The **GTP-protein complex** produces the effect
4. ★ **Intrinsic GTPase** converts **GTP → GDP**, restoring the **inactive resting state** and **terminating** the signal

★★ Heterotrimeric G-proteins (Slide 10):

- **Couple the surface GPCR** to an **ion channel** or an **intracellular catalytic unit** that makes the **second messenger**
- ★ **> 1000 receptors** use them; each GPCR has ★ **7 transmembrane loops** (📝 underlined)
- ★ **Amplification**: one active GPCR → **1, 10 or more** active G-proteins
- ★ **3 subunits: α , β , γ**
- **α subunits** fall into **3 subfamilies: s, i, q**; one GPCR can couple to one or more families



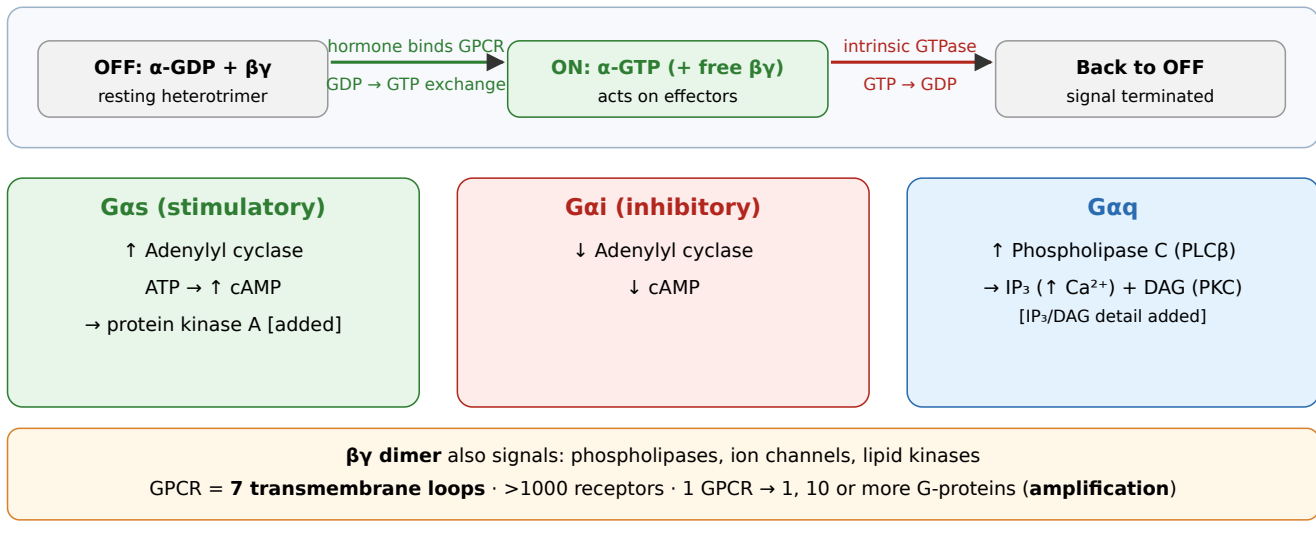
Slide 10: GPCR with 7 transmembrane loops and the α - β - γ G-protein

★★★★ Downstream effectors (Slide 11):

α subunit	Effector	Result
★ Gαs	Stimulates adenylyl cyclase	★ ↑ cAMP (📝 "ATP → cAMP")
★ Gαi	Inhibits adenylyl cyclase	★ ↓ cAMP
★ Gαq	Activates phospholipase C (PLCβ)	[added: → IP $_3$ (↑ Ca $^{2+}$) + DAG (→ PKC)]
G$\beta\gamma$ dimer	Activates phospholipases, ion channels, lipid kinases	

Mnemonic [added]: **s** = stimulate, **i** = inhibit (both on adenylyl cyclase/cAMP); **q** = the odd one → PLC.

GPCR signalling: the G-protein ON/OFF switch and its three α families



Added diagram: the G-protein switch and the Gs / Gi / Gq families

[added examples: Gs, β -adrenergic, glucagon, TSH, ACTH, ADH-V2; Gi, α 2-adrenergic, M2 muscarinic; Gq, α 1-adrenergic, M1/M3 muscarinic, ADH-V1, angiotensin II.]

Recap: GPCRs. 7-loop receptor. GDP $\rightarrow$ GTP turns the G-protein on; intrinsic GTPase turns it off. α , β , γ subunits; one receptor activates many G-proteins (amplification). Gs $\uparrow$ cAMP, Gi $\downarrow$ cAMP, Gq $\rightarrow$ PLC; $\beta\gamma$ also signals.

Link: Some receptors skip the G-protein and are enzymes themselves, or grab an enzyme directly.

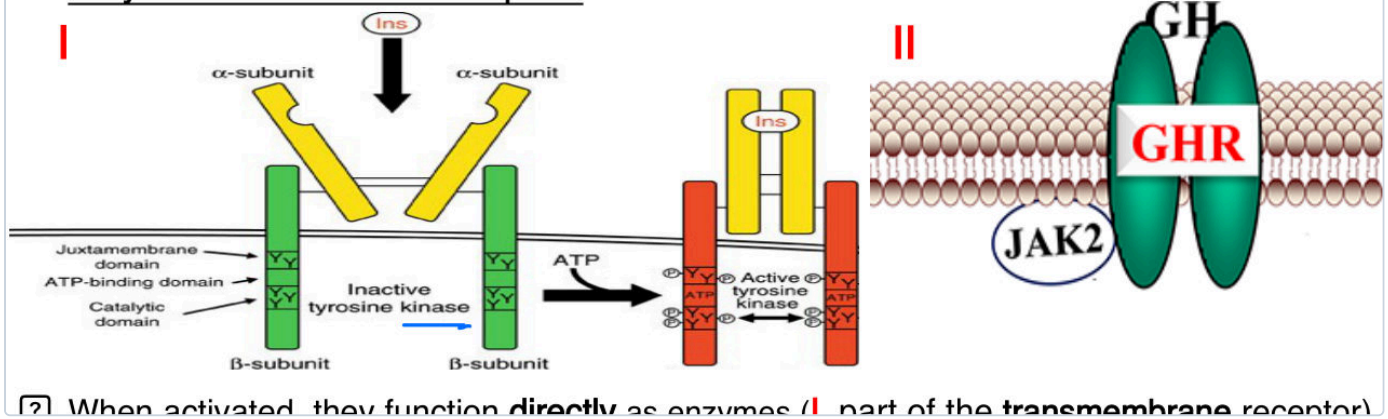
Up next: (C) Enzyme-linked receptors.

2.5 (C) Enzyme-Linked Hormone Receptors (Slide 12)

★ **Hormone-binding site outside; catalytic or enzyme-binding site inside.**

	★ Type I: intrinsic enzyme	★ Type II: enzyme-associated
Enzyme	Part of the transmembrane receptor ("inherited")	The H-R complex activates a separate cytoplasmic enzyme
Inside site	Catalytic domain	Enzyme-binding domain
Slide example	★ Insulin receptor: α -subunits bind insulin; β -subunits have tyrosine kinase (ATP $\rightarrow$ autophosphorylation, inactive $\rightarrow$ active)	★ Growth hormone receptor (GHR) with JAK2

C. Enzyme-Linked Hormone Receptors:



Slide 12: Type I (insulin receptor tyrosine kinase) vs type II (GH receptor + JAK2)

Recap: Enzyme-linked. Type I receptor *is* the enzyme (insulin → tyrosine kinase). Type II recruits a cytoplasmic enzyme (GH → JAK2).

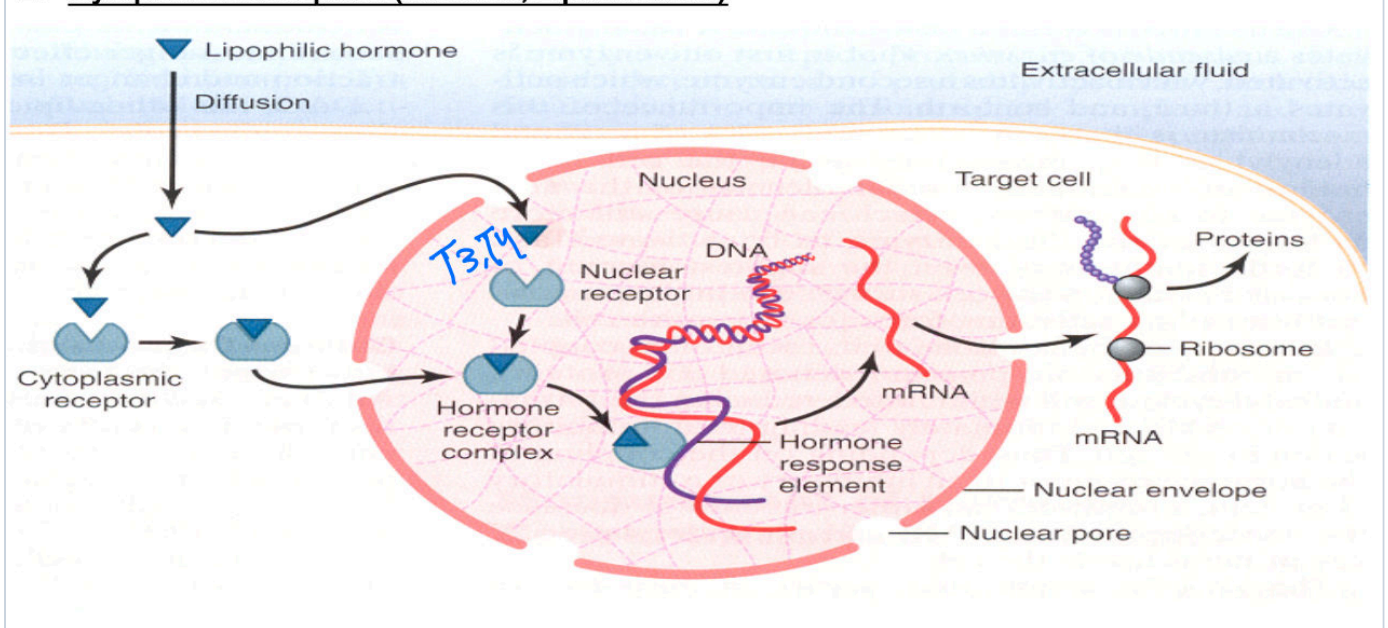
Link: Everything so far serves hormones that stay outside the cell. Lipid-soluble hormones walk straight in.

Up next: Cytoplasmic/nuclear receptors and transcription.

2.6 Cytoplasmic (& Nuclear) Receptors: Transcription (Slides 13-14)

- ★ For **lipid-soluble** hormones (**steroids**), which **cross the membrane** and bind **cytoplasmic (or nuclear)** receptors
- 📝 The lecturer wrote "**T3, T4**" beside the **nuclear receptor** on the figure (thyroid hormones bind in the nucleus, as in Lecture 05)
- ★ The activated **hormone-receptor (H-R) complex** binds the ★ **hormone response element (HRE)** in a gene's regulatory sequence (**promoter**) → alters **transcription and mRNA** → new **proteins**

2. Cytoplasmic receptors (Steroids, Lipid-soluble)



Slide 13: Lipophilic hormone → cytoplasmic or nuclear receptor → HRE → mRNA → protein

Three pathways by which a first messenger alters transcription (Slide 14), 📝 all three scribbled out:

1. Activation of a cytoplasmic latent transcription factor
2. Activation of a cytoplasmic protein kinase cascade → migrates to the nucleus → phosphorylates a nuclear latent transcription factor
3. Binding to a cytoplasmic or nuclear receptor with different effects

[?] There are 3 distinct pathways for 1ry messenger to alter transcription:

1. Activation of **cytoplasmic** latent transcription factor.
2. Activation of **cytoplasmic** protein kinase cascade → migrate to the **nucleus** → phosphorylate a **nuclear** latent transcription factor.
3. Binding to cytoplasmic or nuclear receptor with **different** effects.

Slide 14: The three transcription pathways, crossed out by the lecturer

Recap: Transcription. Steroids (cytoplasm) and T3/T4 (nucleus) form an H-R complex that binds the HRE in the promoter → mRNA → protein. The detailed 3-pathway list was crossed out.

2.7 Conclusion (Slide 16)

- Receptors may be **transmembrane, cytoplasmic or nuclear**
- Transmembrane receptors act via **ion channels, GPCRs or enzyme-linked receptors**
- Ion channels have **different configurations** (pentamer vs tetramer)
- GPCRs are classified by G-protein into **small and heterotrimeric** (*only heterotrimeric is taught on the slides*)
- Hormone-activated enzymes may be **part of the receptor or cytoplasmic**

3. High-Yield Comparison Tables

Table A: Receptor classes

	Ion channel	GPCR	Enzyme-linked	Cytoplasmic/nuclear
Location	Membrane	Membrane	Membrane	Cytoplasm / nucleus
Hormone type	Water-soluble	Water-soluble	Water-soluble	Lipid-soluble (steroids, T3/T4)
Structure	Pentamer (1 pore) / tetramer (4 pores)	7 TM loops + αβγ G-protein	Outside binding site, inside catalytic/binding site	H-R complex + HRE
Example	ACh, GABA; ADH (aquaporin)	Many (> 1000)	Insulin (type I), GH-JAK2 (type II)	Cortisol, T3/T4
Speed [added]	Fastest (ms)	Seconds-minutes	Minutes	Hours

Table B: Gs vs Gi vs Gq (see §2.4)

Table C: Type I vs type II enzyme-linked (see §2.5)

4. Rules, Exceptions, Patterns

- **Water-soluble** → **transmembrane receptor**; **lipid-soluble** → **cytoplasmic/nuclear**.
- **Pentamer** = **one shared pore**; **aquaporin tetramer** = **one pore per subunit**.
- **GDP = off, GTP = on**; the G-protein switches itself off with its **own GTPase**.
- **Every GPCR has 7 transmembrane loops**.
- **Gs ↑ cAMP, Gi ↓ cAMP, Gq → PLC**.
- **βγ is not just a passive partner**: it activates phospholipases, channels, lipid kinases.
- **Type I enzyme-linked = receptor is the enzyme (insulin)**; **type II = recruits one (GH-JAK2)**.
- **Steroid/thyroid H-R complex acts on the HRE in the promoter**.

5. Clinical Correlations *[added; the lecture's only clinical link is the case]*

- **Snake α-neurotoxins / myasthenia gravis** target the nicotinic ACh receptor (pentameric ligand-gated channel).
- **Benzodiazepines/barbiturates** enhance GABA-A (pentameric Cl⁻ channel).
- **Cholera toxin** locks **Gas** on → ↑ cAMP → secretory diarrhoea. **Pertussis toxin** blocks **Gai**.
- **Nephrogenic diabetes insipidus**: defective V2 receptor or aquaporin-2.
- **Type 2 diabetes**: impaired insulin-receptor tyrosine-kinase signalling (insulin resistance).
- **Tyrosine kinase inhibitors** (imatinib) target enzyme-linked signalling in cancer; **JAK inhibitors** (tofacitinib) target type II signalling.

6. Rapid Revision (Last-Day)

- Tree: **transmembrane** (channels, G-protein, enzyme) · **cytoplasmic** (enzyme, transcription). Gives **amplification, distribution, regulation, feedback**.
- Transmembrane receptors for **lipid-insoluble**: peptides, **catecholamines, serotonin, melatonin**; act **directly or via G-proteins**.
- Channels: **pentamer, 5 subunits, 1 pore (ACh, GABA)** · **aquaporin tetramer, 1 pore per subunit (ADH)**.
- GPCR: **GDP → GTP = on; intrinsic GTPase → GDP = off. 7 TM loops, > 1000 receptors, α β γ, 1 → 10+ G-proteins (amplification)**.
- **Gas ↑ AC → ↑ cAMP** · **Gai ↓ AC → ↓ cAMP** · **Gαq → PLC. βγ → phospholipases, channels, lipid kinases**.
- Enzyme-linked: **type I intrinsic (insulin, tyrosine kinase)** · **type II associated (GH + JAK2)**; binding site outside, catalytic/binding site inside.
- Lipid-soluble (steroids; T3/T4 nuclear) → **H-R complex → HRE in promoter → transcription → mRNA → protein**.

7. Exam-Style Questions

Q1 (MCQ, lecture case, Slide 17). A snake toxin irreversibly locks open the **nicotinic acetylcholine receptor** at the neuromuscular junction. This receptor's structural class? **A. Pentameric ligand-gated ion channel** B. Heterotrimeric GPCR C. Enzyme-linked receptor with

intrinsic catalytic activity E. Cytoplasmic (nuclear) receptor **Answer: A.** Slide 8 names ACh (and GABA) receptors as pentameric ligand-gated channels. *[The slide doesn't mark an answer and skips option "D".]*

Q2 (MCQ). What terminates G-protein signalling? a) GDP → GTP exchange **b) Intrinsic GTPase activity of the α subunit** c) Binding of $\beta\gamma$ d) Adenylyl cyclase **Answer: b.** GTP → GDP restores the resting state.

Q3 (MCQ). A hormone increases intracellular cAMP. Its receptor most likely couples to: **a) Gas** b) $G_{\alpha i}$ c) $G_{\alpha q}$ d) JAK2 **Answer: a.**

Q4 (MCQ). Which is a type II (enzyme-associated) receptor? a) Insulin receptor **b) Growth hormone receptor** c) Nicotinic receptor d) Aquaporin **Answer: b.** GHR activates cytoplasmic JAK2; the insulin receptor has its own tyrosine kinase (type I).

Q5 (Short answer). Contrast the pore structure of the nicotinic ACh receptor and aquaporin. **Answer:** ACh receptor: **pentamer**, 5 subunits forming **one** central pore. Aquaporin: **tetramer** with **one pore in each subunit** (4 pores), regulated by ADH.

8. Exam Pitfalls & Traps

1. **GTP = off.** No, **GTP = on**, GDP = off.
2. **Gi activates PLC.** No: Gi **inhibits adenylyl cyclase**; **Gq** activates PLC.
3. **Aquaporin has one central pore.** No, **one per subunit**; the pentamer has one shared pore.
4. **Insulin receptor recruits JAK2.** No, it **is** a tyrosine kinase (type I); GH uses JAK2 (type II).
5. **Catecholamines enter the cell.** No, they are lipid-insoluble → membrane receptors (like serotonin and melatonin).
6. **$\beta\gamma$ is inactive.** It activates phospholipases, channels and lipid kinases.
7. **⚠️ Crossed-out content:** the 3 transcription pathways (Slide 14) were scribbled out; likely not examinable.
8. **⚠️ "Small G-proteins"** appear in the conclusion but aren't taught on any slide.
9. **⚠️ Slide wording:** "cation (K^+)" for ligand-gated channels; the nicotinic receptor mainly carries **Na^+** inward *[added]*. The case's option list jumps from C to E (no D).